

A smaller ABC hit

Disproof of Conjecture 00000008178

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4 October 2026

Result. The triple $(3, 125, 128)$ disproves the assertion that $(2, 3^{10} \cdot 109, 23^5)$ is the smallest ABC hit. The witness also obeys the power cutoff with the explicit positive parameter $\varepsilon = 1/4$.

1. The assertion being disproved

The English and Chinese versions of the conjecture both assert that the smallest ABC hit is the displayed classical triple. The English specifies “the smallest $a+b=c$ ”. Thus minimality is measured by $c=a+b$. The conjecture also makes counting-density and algorithm-complexity assertions. Its assertions are conjunctive; disproving the minimum assertion disproves the conjecture, without deciding those other assertions.

For a positive integer n , let

$$\text{rad}(n) = \prod_{\substack{p|n \\ p \text{ prime}}} p.$$

An ABC triple consists of positive integers $a < b$ and $c = a + b$ that are pairwise coprime. An ABC hit satisfies $\text{rad}(abc) < c$. To check the conjecture’s additional power-cutoff wording as well, we impose the stronger condition

$$\text{rad}(abc) \leq c^{1-\varepsilon}, \quad 0 < \varepsilon < 1. \quad (1)$$

This explicit check avoids relying solely on an ordinary hit with a radical barely below c . If the schematic cutoff $\text{Rad}(n) \leq n^{1-\varepsilon}$ uses $n = abc$, condition (1) implies it too, because $abc \geq c$ and $1 - \varepsilon > 0$.

2. Exact counterexample

Take $a = 3$, $b = 125 = 5^3$, and $c = 128 = 2^7$. Then

$$3 + 125 = 128, \quad \gcd(3, 125) = \gcd(3, 128) = \gcd(125, 128) = 1.$$

Their product is $48000 = 2^7 \cdot 3 \cdot 5^3$, so its set of prime divisors is exactly $\{2, 3, 5\}$ and

$$\text{rad}(3 \cdot 125 \cdot 128) = 2 \cdot 3 \cdot 5 = 30 < 128.$$

All entries exceed 1; the counterexample therefore does not depend on whether an optional convention excludes $a = 1$.

For $\varepsilon = 1/4$, exact integer arithmetic gives

$$30^4 = 810000 < 2097152 = 128^3.$$

Since the fourth-power function is strictly increasing on the nonnegative real numbers, this yields $30 < 128^{3/4}$. In particular,

$$\text{rad}(3 \cdot 125 \cdot 128) \leq 128^{1-1/4}.$$

On the other hand, the proposed smallest triple has

$$2 + 3^{10} \cdot 109 = 2 + 6436341 = 6436343 = 23^5,$$

and $128 < 6436343$. A smallest hit with this value of c would require $23^5 \leq 128$ when applied to our witness, a contradiction. The same contradiction applies to a proposed minimum in the fixed $\varepsilon = 1/4$ cutoff class. This proves the claimed minimality false. $\square$

3. Kernel-checked formalization

The accompanying Lean 4.19.0 project pins Mathlib to commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

It defines the radical using `Nat.primeFactors`, the actual finite set of prime divisors, and its product. It proves the exact set $\{2, 3, 5\}$ from the prime-power factorization. Positivity, addition, pairwise coprimality, and the strict ordering of the two values of c are likewise checked rather than assumed.

The power cutoff uses Mathlib's real numbers and real exponentiation. The proof applies `Real.rpow_le_rpow_iff` with exponent 4, combines powers using `Real.rpow_mul`, and reduces to exact arithmetic. No numerical approximation or floating-point computation is used.

The principal declarations in `lean/Main.lean` are:

- `witness_primeFactors` and `witness_radical`: the actual prime divisors and radical of abc ;
- `witness_cutoff`: the full ABC hit and real power bound;
- `not_smallest_hit` and `not_smallest_cutoff_hit`: the two minimality negations;
- `counterexample`: the combined result with $0 < \varepsilon < 1$, the explicit witness, and both negations.

The project contains no additional axioms, incomplete proofs, or `native_decide`. The axiom audit and reproduction commands are recorded in `VERIFICATION.md`. No auxiliary numerical program is needed for the proof.